

Course Objectives:

Learn the fundamentals of designing and developing modern, responsive, and interactive user interfaces for web applications.

Course Content:

UNIT 1: Introduction to Frontend Development

- Introduction to Web Development
- Role of Frontend Development
- Client-Side vs Server-Side
- Web Browsers and Developer Tools
- Introduction to HTML, CSS, and JavaScript

UNIT 2: HTML and CSS

- HTML Document Structure
- Semantic HTML
- Forms and Input Elements
- CSS Selectors and Properties
- Flexbox and CSS Grid
- Responsive Web Design

UNIT 3: JavaScript Fundamentals

- Variables and Data Types
- Functions and Scope
- Arrays and Objects
- DOM Manipulation
- Events and Event Handling
- ES6 Features

UNIT 4: Modern Frontend Frameworks

- Introduction to React
- Components and Props
- State Management
- Hooks
- Routing
- API Integration

UNIT 5: Advanced Frontend Development

- Responsive and Mobile-First Design
- State Management
- Authentication and Authorization
- Performance Optimization
- Testing and Debugging
- Deployment of Frontend Applications

Suggested Books:

- Jon Duckett — HTML and CSS: Design and Build Websites
- Marijn Haverbeke — Eloquent JavaScript
- Alex Banks and Eve Porcello — Learning React

Course Outcomes:

- Understand the fundamentals of frontend web development.
- Build responsive and user-friendly web interfaces.
- Use HTML, CSS, and JavaScript effectively.
- Develop interactive applications using React.
- Integrate frontend applications with backend APIs.